

王胤

CONTACT INFORMATION

王胤
普陀区同普路 688 弄 20 号 1701
上海, 200333

Phone: 13916006420
E-mail: alexwy@gmail.com
Web: <http://web.eecs.umich.edu/~yinw/>

WORK EXPERIENCE

VP Engineering & CTO backup, 饿了么, 上海

02/2017–now

基础设施部门负责人 (160 人)

- 引入硅谷管理思想, 改进绩效考核系统, 打造工程师文化, 将团队从被动接需求向自驱动转变
- 带领运维和测试团队转型, 一年时间将部门开发: 非开发比例从 4:6 改变到 6:4
- 引入国外技术开展创新, 帮助业务。比如, 开发了一个高压缩比的内存数据库, 同时提供监控系统的实时性加上 Hive 的大数据分析能力, 性能测试十台机器 2s 分析 5 亿宽表数据。目前已经开始向业务部门推广
- 引入国外技术提升机器利用率, 除了常见的容器, 资源池化, 分布式存储系统等, 还有异步服务系统, 替代 redis + mysql 架构的 write-through cache 存储服务
- 引入国外理念与技术提升研发团队工程效率和产出, 比如开发团队自测自运维, 统一代码仓库, 代码 review 机制, 单机测试环境, 自动测试, 持续集成和发布等

物流研发部门负责人 (130 人)

- 积极尝试打破业务研发“来料加工”的被动局面, 向硅谷全能型开发看齐。带领研发团队走近业务, 从自主承接简单需求开始, 逐步承担更多责任, 提升迭代效率。同时将产品经理从繁琐 PRD 和项目管理等任务中解放出来, 更多思考产品和业务方向
- 引导研发部门进行业务创新 (细节不便透露)

Senior Staff Software Engineer, Facebook

05/2013–02/2017

Maps, Founder and Tech Lead

- 利用深度学习等智能技术, 带领不到 50 人的团队提供了全栈的地图相关服务, 包括地图数据, 地址编码反编码, LBS 服务, 移动端定位服务, 室内定位等。实现了 Facebook 地图从第三方依赖到完全独立的转型, 地图部门因此成立
- 利用深度学习提取高分辨率卫星图像中的道路和建筑信息, 不到 20 人运营团队完成 OpenStreetMap 泰国地图。自动提取的建筑数据比手工绘制商业数据库覆盖率高一倍, 准确率相当。是亿万美金广告产品 Store Visit 的基石
- 自研 LBS 服务系统 50 台机器提供千万级的 qps, 比 Boost 和 Google S2 快 100 倍, 在 Mark Q&A 公司全体会议中作为 fix-of-the-week 重点表扬
- 利用 RocksDB 引擎开发了一个地图渲染工具, 迅速定制各种地图渲染需求

Facebook AI Research (FAIR)

- 深度学习基础设施建设, 性能优化
- 卫星图像处理, 文字识别, 视频分析处理

Data Infrastructure

- 开发了一个针对非 HDFS 系统的 MapReduce 任务调度管理系统, 比如可以直接在 OLTP 系统上进行 MapReduce 任务。是 Facebook 对 MySQL, 消息数据, 照片等在线数据进行批处理任务的唯一工具 [USENIX ATC '15], 开源地址: <https://facebook.github.io/bistro/>

Senior Research Scientist, HP Labs, Palo Alto
Big Data Lab

05/2008–04/2013

- 领导地理信息大数据平台研究，开发了 CrowdAtlas 项目利用 GPS 数据自动更新地图，获得 ACM MobiSys 最佳 demo 奖，并得到 Apple CEO Tim Cook 重视

Operating System Lab

- 利用控制理论自动发现并解决软件 bug，在系统领域顶级会议比如 OSDI, POPL, EuroSys 等发表多篇论文并获奖，同时在 HP Enterprise 工作流系统中应用

Internships During Graduate School

IBM Research, Almaden summer 2007
log and performance data mining for enterprise storage management systems [MASCOTS '08]

Merit Networks Inc., Ann Arbor summer 2006
machine learning for decentralized root-cause diagnosis of network routing anomalies using BGP logs

HP Labs., Palo Alto summer 2005
Designed and implemented distributed fair scheduling framework for a distributed storage system. Developed algorithms that provably optimize a variety of fairness objectives [FAST '07]

Microsoft, Shanghai summer 2002
Led a test team on a commercial bug management product. Verified testability of specifications, designed test cases in parallel with product development.

EDUCATION	Ph.D., Computer Science and Engineering, U. Michigan Ann Arbor 2008 Dissertation: “Software Failure Avoidance Using Discrete Control Theory” Advisors: Scott Mahlke, Stéphane Lafortune
-----------	--

M.Eng., Electrical Engineering, Shanghai Jiao Tong Univ. 2003 Thesis: “Structural Control of Resource Allocation Systems Using Model Checking”
--

B.Eng., Electrical Engineering, Shanghai Jiao Tong Univ. 2000
--

EXTERNAL TALKS	“硅谷管理模式在本地企业的实践” 2017 CSDI, Guangzhou
-------------------	---

“Scaling Maps at Facebook” 2016 keynote, ACM SIGSPATIAL conference, San Francisco

“How can AI help us make maps” 2016 State of the Map US, Seattle
--

“Facebook Location Infrastructure” 2015 Top 100 Summit, Beijing

“GeoSpatial Features in RocksDB” 2014 Data @Scale, Boston

	“Bistro: Scheduling Data-Parallel Jobs against Live Production Systems” Top 100 summit, Beijing	2014
	20+ more academic conference talks	before 2014
HONORS AND AWARDS	Chinese Thousand Young Talent Award 10 recipients in CS area for all oversea Chinese under 40	2015
	ACM SIGSPATIAL Cup finalists three times in a row (3rd, 1st, 4th) a programming competition with 30+ participating teams worldwide each year	2012-2014
	CrowdAtlas demo received ACM Mobisys '13 best video award vimeo.com/alexwy/crowdatlas	2013
	POPL '09 paper nominated by ACM SIGPLAN for CACM Research Highlights article 2009 only 2 of all 2009 SIGPLAN conference papers were nominated	
	OSDI '08 paper featured in <i>IEEE Computer</i> news briefs, vol. 42, No. 8, pp. 16-18	2008
	Dissertation received Distinguished Achievement Award from U. Michigan College of Engineering ACM Doctoral Dissertation Award nomination from U. Michigan	2008
	Fellowships, Department of EECS, Univ. of Michigan	2008 & 2004
	Fellowship, HP Labs	2006
	Outstanding Performance Internship, Microsoft (Shanghai) Inc.	2002
	Excellent Student Award, Shanghai Jiaotong University (ranked 3/108)	1997
	Several 1st Class Awards from National Math Olympic Competitions	before 1996
SELECTED PUBLICATIONS	Conferences: “Bistro: Scheduling Data-Parallel Jobs Against Live Production Systems” A. Goder, A. Spiridonov, Y. Wang ATC 2015: Usenix Annual Technical Conference “Greedy Map Generalization by Iterative Point Removal” Y. Chen, Y. Wang , R. Chen, H. Chen, B. Zang GIS 2014: ACM International Conference on Advances in Geographic Information Systems “Point-Polygon Topological Relationship Query Using Hierarchical Indices” T. Zhou, H. Wei, H. Zhang, Y. Wang , Y. Zhu, H. Guan, H. Chen GIS 2013: ACM International Conference on Advances in Geographic Information Systems “CrowdAtlas: Self-Updating Maps for Cloud and Personal Use” Y. Wang , X. Liu, H. Wei, G. Forman, C. Chen, Y. Zhu MobiSys 2013: ACM Conference on Mobile Systems, Applications and Services	

“Fast Viterbi Map Matching with Tunable Weight Functions”

H. Wei, **Y. Wang**, G. Forman, Y. Zhu, H. Guan

GIS 2012: ACM International Conference on Advances in Geographic Information Systems

“Mining Large-Scale, Sparse GPS Traces for Map Inference: Comparison of Approaches”

X. Liu, J. Biagioni, J. Eriksson, **Y. Wang**, G. Forman, Y. Zhu

KDD 2012: ACM Conference on Knowledge Discovery and Data Mining

“The Theory of Deadlock Avoidance via Discrete Control”

Y. Wang, S. Lafortune, T. Kelly, M. Kudlur, S. Mahlke

POPL 2009: ACM Symposium on Principles of Programming Languages

“Gadara: Dynamic Deadlock Avoidance for Multithreaded Programs”

Y. Wang, T. Kelly, M. Kudlur, S. Lafortune, S. Mahlke

OSDI 2008: USENIX Symposium on Operating Systems Design and Implementation

“Discrete Control for Safe Execution of IT Automation Workflows”

Y. Wang, T. Kelly, S. Lafortune

EuroSys 2007: ACM European Conference on Computer Systems

“Proportional-Share Scheduling for Distributed Storage Systems”

Y. Wang, A. Merchant

FAST 2007: USENIX Conference on File and Storage Technologies

Journals:

“Eliminating Concurrency Bugs in Multithreaded Software: A New Approach Based on Discrete-Event Control”

H. Liao, **Y. Wang**, J. Stanley, S. Lafortune, S. Reveliotis, T. Kelly, S. Mahlke

IEEE Trans. on Control Systems Technology, vol. 21, no. 6, pp. 2067–2082, 2013

“Optimal Liveness-Enforcing Control of a Class of Petri Nets Arising in Multithreaded Software”

H. Liao, S. Lafortune, S. Reveliotis, **Y. Wang**, S. Mahlke

IEEE Trans. on Automatic Control, vol. 58, no. 2, pp. 1123–1138, 2013

“Concurrency Bugs in Multithreaded Software: Modeling and Analysis Using Petri Nets”

H. Liao, **Y. Wang**, H.K. Cho, J. Stanley, T. Kelly, S. Lafortune, S. Mahlke, S. Reveliotis

Journal of Discrete Event Dynamic Systems, vol. 23, no. 2, pp. 157–195, 2013

“Designing Compact and Maximally Permissive Deadlock Avoidance Policies”

A. Nazeem, S. Reveliotis, **Y. Wang**, S. Lafortune

IEEE Trans. on Automatic Control, vol. 56, no. 8, pp. 1818–1833, 2011

“Eliminating Concurrency Bugs with Control Engineering”

T. Kelly, **Y. Wang**, S. Lafortune, S. Mahlke

IEEE Computer, vol. 42, no. 12, pp. 52–60, Dec. 2009

“Diagnosis of Discrete Event Systems Using Decentralized Architectures” (invited paper)

Y. Wang, T. Yoo, S. Lafortune

U.S. PATENT	“System and Method for Providing, by a Plurality of Schedulers, Differentiated Service to Consumers of Distributed Resources,” US Patent 7823154, Y. Wang , A. Merchant	2010
	“Middleware for Extracting Aggregation Statistics to Enable Light-weight Management Planners,” US Patent, 8745637, A. Singh, S. Uttamchandani, Y. Wang	2011
	“Composition of Services”, US Patent, 9009281 Y. Wang , H. Nezhad, R. Swaminathan, S. Singhal	2013
	“Providing Goods or Services”, US Patent application, 20130111430 Y. Wang , A. Nazeem, R. Swaminathan	2013
	“Identifying Impact of a Traffic Incident on a Road Network” US Patent, 9047495, M. Miller, C. Gupta, and Y. Wang	2013
	“Road Maps from Clusters of Line Segments of Multiple Sources” US Patent application, 20140095062, Y. Wang , X. Liu, and Y. Zhu	2014
	“Generating A Non-Deterministic Model of a Process for a Goal” US Patent application, 20140236554, Y. Wang , and A. Nazeem	2014
	“Updating Road Maps” US Patent application, 20140278055, Y. Wang , G. Forman, and X. Liu	2014
	“Map Matching” US Patent application, 20150354973, Y. Wang , H. Wei, and G. Forman	2013
	“Systems and Methods for Providing Quantitative Evaluation of Map Quality ” US Patent application, 20170140218, Y. Wang	2015
ACADEMIC ACTIVITIES	Academic Positions:	
	Professor, CS Department, Tongji University	2016–Present
	Visiting Scholar, EECS Department, University of Michigan	2009–2015
	Ph.D Thesis Committee:	
	Hongwei Liao, University of Michigan	2012
	Conference Organizer:	
	“Real-World Challenges for Data Stream Mining,” ECML/PKDD 2013 workshop	
	“Modeling, Control, and Analysis of Software Systems,” CDC 2012 invited session	
	“Controlling Software Execution,” ACC 2012 workshop	
	Program Committee:	
	WODES 2010, 2012, 2014 (Workshop on Discrete Event Systems)	
	FeBID 2010, 2011 (Control Engineering workshop)	
	ICEBE 2010, 2011	
	IMIS 2011	
	External Reviewers:	

Conferences: OSDI 2008, SIGMETRICS 2006, FAST 2007, EuroSys 2008, 2012,
 USENIX Annual Tech 2007, 2014,
 Journals: *IEEE Trans. on Automatic Control*, *J. Discrete Event Dynamic Systems*,
IEEE Trans. on Industrial Informatics, *IEEE Trans. on Systems, Man and Cybernet-*
ics,
ACM Operating Systems Review, *ACM Trans. on Programming Languages and Sys-*
tems,
Service Oriented Computing and Applications, *Automated Software Engineering Jour-*
nal,

Senior members, ACM & IEEE

Funding:

Co-authored \$500K NSF project proposal (UMich)	2008
Co-authored \$225K HP Labs Open Innovation proposal (UMich)	2007–10
Co-authored \$225K HP Labs Open Innovation proposal (SJTU)	2011-14

TEACHING
EXPERIENCE

Guest Lecture:	
SJTU CS: Computer Architecture	Spring 2013
UMich EECS 598: Ubiquitous Parallelism	Fall 2010
Graduate Student Instructor:	
UMich EECS 306: Signal and Systems II	Fall 2005
UMich EECS 206: Signal and Systems I	Fall 2004, Winter 2005

REFERENCES

George Forman (Senior Research Scientist, HP Labs)
george.forman@hp.com

Norman P. Jouppi (Distinguished Engineer, Google)
jouppi@acm.org

Matt Welsh (Senior Staff Software Engineer, Google)
mdw@mdw.la

Arif Merchant (Staff Research Scientist, Google)
aamerchant@google.com

Robert E. Tarjan (James S. McDonnell Distinguished Professor, Princeton)
ret@cs.princeton.edu

Edward A. Lee (Robert Pepper Distinguished Professor, Berkeley)
eal@eecs.berkeley.edu